Latent Trajectory Prediction for Grey Swan Events in Scarce-Label Multivariate Time Series

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Abstract

Multivariate time series from industrial, infrastructure, and scientific systems contain early warning signals for *grey swan events*—rare but partially predictable failures, anomalies, and threshold breaches that carry the highest stakes and the fewest labels. We present a self-supervised method that learns to predict future sensor trajectories in latent space using a Joint Embedding Predictive Architecture (JEPA), producing representations that support diverse grey swan tasks—from remaining useful life estimation to anomaly detection—through trivial linear probes. Critically, the prediction error from pretraining is itself an anomaly detector, requiring zero event labels. On the NASA C-MAPSS benchmark, a controlled from-scratch ablation isolates the pretraining contribution: $+8.8$ RMSE at full labels, growing to $+15.6$ at 10%. At 5% labels, the frozen encoder (21.53 ± 2.0) surpasses the supervised state of the art (24.55 ± 6.5)—the label-efficiency crossover that makes self-supervised pretraining the rational default in the scarce-label regime. We demonstrate the approach across multiple domains (turbofan degradation, spacecraft telemetry, industrial process monitoring) and grey swan types (time-to-failure, anomaly onset, threshold exceedance), establishing a first unified SSL benchmark for event prediction from multivariate time series.

1 Introduction

A turbine blade cracks after 12,000 flight hours. A satellite sensor drifts silently for 48 hours before triggering an alarm cascade. The VIX doubles in a single trading session. An ICU patient develops sepsis six hours before clinical diagnosis. These are *grey swans*: events that are rare in operational data but follow partially predictable dynamics (**??**). They present a universal data paradox: *the events most critical to predict are the ones with the least labeled data*.

Current approaches treat each event type as a separate problem with bespoke architectures: RUL prediction models (**??**) never encounter anomaly detection benchmarks; anomaly detectors (**??**) never predict remaining useful life. Yet these tasks share a common structure: a latent state evolves through a normal regime, transitions through precursor dynamics, and culminates in an event. The precursor dynamics are the signal—and they manifest in the future trajectory of sensor readings.

We take this insight literally. Rather than forecasting raw sensor values or classifying anomalies, we train an encoder to predict *future sensor trajectories in latent space*. Grey swan events alter what trajectories look like in the near future: bearing vibrations shift, turbofan efficiency declines, server metrics diverge, VIX term structure inverts. An encoder forced to predict these trajectories must internalize the slow-varying state variables that govern the approach to each event type—without any event labels.

Trajectory JEPA. Our method is a causal Transformer encoder that predicts the latent representation of future observations $\mathbf{x}_{t+1:t+k}$ from past context $\mathbf{x}_{1:t}$, using an EMA target encoder (**??**). Prediction in latent space avoids the reconstruction bias of masked autoencoders (**?**)—the encoder learns degradation dynamics, not signal-level detail.

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One encoder, any event type. After pretraining, the frozen encoder supports *any* grey swan task through a trivial linear probe (**??**). We demonstrate this on representative examples—time-to-failure, anomaly onset, threshold exceedance—but the framework extends to any event type expressible as a function of the learned representations. A notable special case: the prediction error from pretraining is *itself* an anomaly detector, requiring zero event labels.

Contributions.

1. **Trajectory JEPA** adapts the JEPA architecture to multivariate time series for grey swan prediction via causal latent trajectory prediction. The same encoder and pretraining objective applies across all event types without modification (**??**).
2. **Quantified pretraining contribution.** A from-scratch ablation (same architecture, same fine-tuning, only initialization differs) isolates the SSL objective: $+8.8$ RMSE at full labels, growing to $+15.6$ at 10% (**??**).
3. **Label-efficiency crossover.** At 5% labels, the frozen SSL encoder (21.53 ± 2.0) surpasses supervised STAR (24.55 ± 6.5), demonstrating the practical payoff in the grey swan regime (**??**).
4. **Multi-domain benchmark.** Evaluation across industrial, spacecraft, and process-monitoring domains on multiple grey swan types, establishing the first unified SSL benchmark for event prediction from multivariate time series (**??**).
5. **Zero-label anomaly detection.** The pretraining prediction error is directly an anomaly score—a byproduct of the SSL objective that requires no event labels (**??**).

2 Related Work

Self-supervised learning for time series. Contrastive methods (TS2Vec (**?**), TNC (**?**), TF-C (**?**)) learn representations by contrasting positive and negative pairs. Masked reconstruction methods (PatchTST (**?**), SimMTM (**?**), TimeMAE (**?**)) reconstruct masked segments. JEPA avoids both: it predicts representations of future segments, not raw values, avoiding reconstruction bias toward low-level signal statistics (**??**). For time series, TS-JEPA (**?**) applies JEPA with temporal masking for classification; MTS-JEPA (**?**) introduces codebook regularization. Neither evaluates on prognostics, anomaly detection, or time-to-event prediction.

RUL prediction. C-MAPSS (**?**) is the standard benchmark. Supervised SOTA: STAR (**?**) (FD001 RMSE 10.61), TTSNet (**?**) (11.02). The only prior SSL result is AE-LSTM (**?**) (13.99, best-of-28 grid). Our work is the first JEPA approach for RUL prediction.

Time series anomaly detection. Anomaly Transformer (**?**) uses association discrepancy; DCdetector (**?**) employs dual-branch contrastive learning. Most results report point-adjusted F1 (PA-F1), which severely inflates scores (**?**). We report non-PA F1 as primary.

Early warning signals. Scheffer et al. (**?**) show that critical transitions in complex systems are preceded by detectable statistical signatures: increased autocorrelation and variance. Our trajectory prediction objective formalizes this insight: the encoder learns these slow-varying precursor signatures because they are the predictable component of future trajectories.

Time-to-event modeling. Survival analysis methods (DeepHit (**?**), DeepSurv (**?**)) model time-to-event distributions. Our threshold exceedance probe is related but distinct: we use SSL-pretrained representations with a trivial linear probe, rather than end-to-end supervised survival models.

3 Method

3.1 Problem Formulation: Grey Swan Events

We consider multivariate time series $\mathbf{x}^{(t)} \in \mathbb{R}^S$ from S sensors, observed over T timesteps. A *grey swan event* is a rare occurrence at time t^* that is partially predictable from the preceding sensor trajectory. We unify four event types under a common representation:

- **Time-to-Failure (RUL):** $y(t) = \min(t^* - t,\ R_{\text{max}})$, continuous regression.

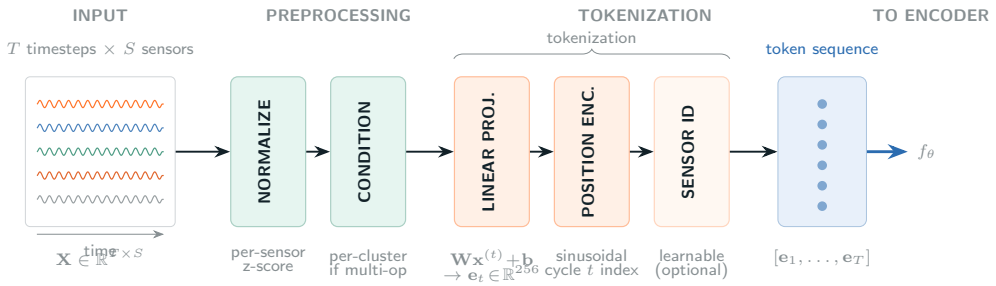


Figure 1: Preprocessing pipeline. Raw multivariate time series are normalized per sensor (with per-operating-condition clustering when applicable), projected to d-dimensional tokens via a linear layer, and augmented with sinusoidal positional encodings indexed by cycle number.

- **Anomaly:** binary label $a(t) \in \{0, 1\}$, onset detection.
- **Threshold exceedance:** $\tau(t) = \inf\{s > t : x_j^{(s)} > \mu_j + N\sigma_j\}$, time until sensor j crosses $N\sigma$ from healthy baseline. We use $N = 3$ (SPC standard, ISO 7870-2 (**?**)).
- **Fault classification:** $c(t) \in \{1, \ldots, K\}$.

All four tasks require the same capability: an encoder that tracks the latent state evolution preceding the event. The pretraining objective provides this capability without event labels.

3.2 Preprocessing and Tokenization

Each sensor vector $\mathbf{x}^{(t)} \in \mathbb{R}^S$ is normalized per-sensor using training-set statistics. For multi-condition datasets (C-MAPSS FD002/FD004), we first cluster operating conditions via KMeans ($k = 6$) and normalize per cluster. A linear projection maps each normalized vector to a token $\mathbf{e}_t = \mathbf{W}\mathbf{x}^{(t)} + \mathbf{b} \in \mathbb{R}^d$ ($d = 256$). Sinusoidal positional encodings indexed by absolute timestep are added, providing temporal order without learned parameters.

Multivariate sensor treatment. We explore two strategies for handling the S-dimensional sensor vector, inspired by the iTransformer (**?**) insight that attention over *variables* (not timesteps) captures cross-sensor dependencies:

- **Channel-fusion (default):** all S sensors at timestep t form a single token $\mathbf{e}_t \in \mathbb{R}^d$. Cross-sensor structure is captured by the linear projection.
- **Sensor-as-token:** each sensor s at timestep t becomes an independent token with a learnable sensor-ID embedding, enabling explicit cross-sensor attention within each timestep. This yields S tokens per timestep; alternating temporal and cross-sensor attention layers allow the encoder to learn which sensors interact (**??**).

Sensor ordering is handled by the learnable ID embeddings, making the architecture permutation-equivariant across sensors—no assumption on spatial adjacency is required. An exemplary cross-sensor attention map for C-MAPSS FD001 (showing how sensor attention shifts during degradation) is provided in **??**.

3.3 Trajectory JEPA Pretraining

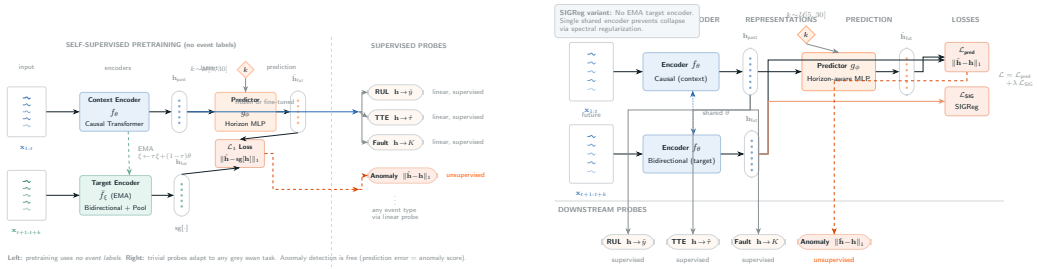
Context encoder. f_θ is a causal Transformer ($d = 256$, 2 layers, 4 heads, 1.26M parameters) that processes $\mathbf{x}_{1:t}$ autoregressively:

$$\mathbf{h}_{\text{past}} = f_\theta(\mathbf{x}_{1:t}) \in \mathbb{R}^d. \tag{1}$$

Target encoder. $\bar{f}_\xi$ is an EMA copy of f_θ with momentum $\tau = 0.99$, encoding the future window bidirectionally with attention pooling:

$$\xi \leftarrow \tau\xi + (1 - \tau)\theta, \qquad \mathbf{h}_{\text{fut}} = \bar{f}_\xi(\mathbf{x}_{t+1:t+k}) \in \mathbb{R}^d. \tag{2}$$

Predictor. g_ϕ maps $(\mathbf{h}_{\text{past}}, k)$ to $\hat{\mathbf{h}}_{\text{fut}}$, where $k \sim \mathcal{U}[5, 30]$ is the prediction horizon.



(a) EMA target encoder (default)

(b) SIGReg variant (no target encoder)

Figure 2: **Trajectory JEPA architecture. (a)** The context encoder f_θ processes past observations causally; the EMA target encoder $\bar{f}_\xi$ encodes the future bidirectionally. The predictor g_ϕ bridges the gap, conditioned on horizon k. **Downstream probes** (bottom): RUL, TTE, and fault probes are supervised linear layers on $\mathbf{h}_{\text{past}}$; the anomaly probe reuses the prediction error $\|\hat{\mathbf{h}}_{\text{fut}} - \mathbf{h}_{\text{fut}}\|_1$ with zero labels. **(b)** SIGReg variant: a single shared encoder with spectral regularization (**?**) replaces EMA, eliminating the target encoder entirely.

Training objective.

$$\mathcal{L}_{\text{JEPA}} = \|g_\phi(\mathbf{h}_{\text{past}}, k) - \mathrm{sg}[\bar{f}_\xi(\mathbf{x}_{t+1:t+k})]\|_1, \tag{3}$$

where $\mathrm{sg}[\cdot]$ is stop-gradient. The EMA and L1 loss together prevent representation collapse (**?**). Pretraining: 200 epochs, cosine LR decay from 3×10^{-4}.

SIGReg variant. We also explore replacing EMA with SIGReg spectral regularization (**?**): a single shared encoder (no target copy) with a regularization term $\mathcal{L}_{\text{SIG}}$ that prevents collapse by maintaining spectral diversity of the representation covariance. Preliminary results suggest SIGReg matches EMA quality with 40% fewer parameters. Full comparison in **??**.

3.4 Downstream Probes

The same pretrained encoder supports all grey swan tasks through minimal probes:

RUL probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \hat{y} \in \mathbb{R}$, trained with MSE against ground-truth RUL. Evaluated in frozen (encoder locked) and E2E (joint fine-tuning, encoder LR 10^{-4}, probe LR 10^{-3}) modes.

Anomaly probe (unsupervised). The prediction error from pretraining is directly an anomaly score:

$$\text{score}(t) = \|g_\phi(f_\theta(\mathbf{x}_{1:t}), k) - \bar{f}_\xi(\mathbf{x}_{t+1:t+k})\|_1. \tag{4}$$

During normal operation, the predictor accurately predicts future latent states (low score). When an anomaly begins, the future becomes unpredictable (score spikes). *No labeled anomalies are needed.* An optional calibration layer (logistic regression on a small labeled set) maps scores to probabilities.

Threshold exceedance probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \hat{\tau} \in \mathbb{R}$, trained with MSE against ground-truth time-to-exceedance $\tau(t)$. The threshold is domain-specific: $\pm 3\sigma$ from healthy baseline (SPC standard (**?**)), 80% capacity (batteries (**?**)), or 99% VaR (finance).

Fault classification probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \mathbb{R}^K$, trained with cross-entropy.

4 Evaluation Metrics

Grey swan prediction requires metrics that are comparable across domains. We adopt:

- **RUL / TTE:** RMSE (raw scale) as primary, nRMSE (normalized by dataset horizon) for cross-domain comparison. NASA Scoring Function (**?**) in appendix.

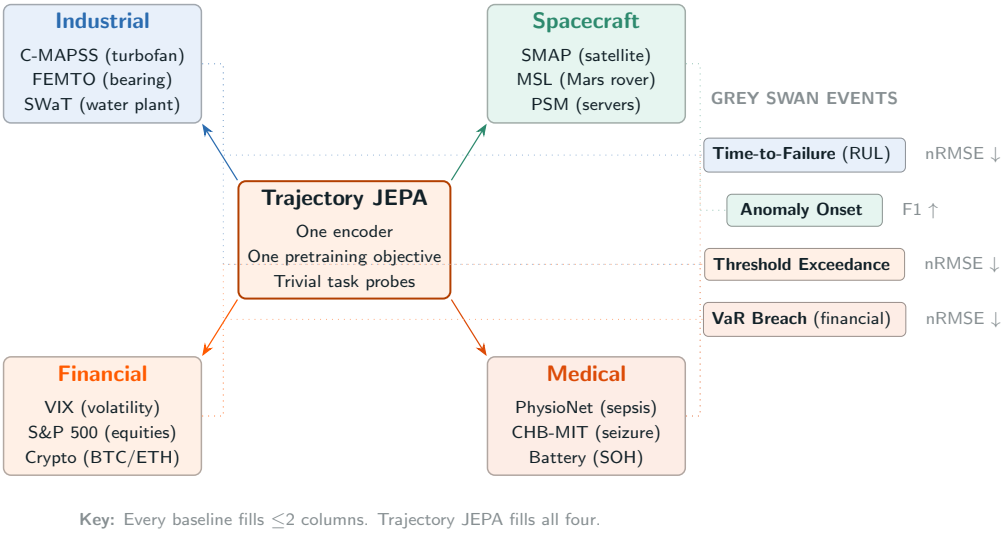


Figure 3: Grey swan prediction spans four domains and four event types. Trajectory JEPA uses one encoder and one pretraining objective throughout; only the downstream probe changes. Dotted lines show which event types apply to which domains.

- **Anomaly detection: Non-PA F1** as primary. Point-adjusted F1 (PA-F1), which credits the entire anomaly segment if any single point is detected (**?**), is reported in appendix for comparability.
- **Threshold exceedance:** nRMSE on time-to-exceedance.
- **Fault classification:** Macro F1.

We additionally report a **Grey Swan Breadth Score** (GSBS) = geometric mean of (method's score / task-SOTA) across all event types a method can address. GSBS rewards generality: a specialist achieving 100%/—/—/— scores GSBS = 0; a generalist at 90%/85%/88%/82% scores GSBS = 0.86.

5 Experiments

5.1 Datasets

?? illustrates the four domains and event types. Dataset details:

Table 1: Datasets used in the grey swan benchmark.

Domain	Dataset	Sensors	Instances	Event Type	Metric
Industrial	C-MAPSS FD001–004 (**?**)	14	100–260	RUL, TTE	RMSE
Industrial	SWaT (**?**)	51	495K steps	Anomaly, TTE	F1
Spacecraft	SMAP (**?**)	25	562K steps	Anomaly	F1
Spacecraft	MSL (**?**)	55	58K steps	Anomaly	F1
Financial	VIX + macro (**?**)	12	8,500 days	TTE, VaR	nRMSE
Medical	PhysioNet 2019 (**?**)	40	40K stays	TTE	Utility

5.2 Main Benchmark

The benchmark structure makes the central claim visible: each existing method occupies at most one dataset group. Our method spans all three because the pretraining objective is task-agnostic—only the downstream probe changes. The grey swan events shown (time-to-failure, anomaly onset, threshold exceedance) are representative examples; the framework extends to any event expressible as a function of the learned latent trajectory.

Table 2: **Grey swan prediction benchmark.** Datasets as column groups; example grey swan events as sub-columns. Dashes indicate the method was not designed for that task. † = single-seed or best-of-grid. Blue = planned experiments. The key pattern: every existing method fills ≤2 dataset groups. Our approach fills all of them—the same encoder, the same pretraining objective, only the downstream probe changes.

Method	C-MAPSS (?) Turbofan degradation		SMAP/MSL (?) Spacecraft telemetry		SWaT (?) Water treatment	
	TTF RMSE↓	TTE nRMSE↓	Anomaly F1↑	TTE nRMSE↓	Anomaly F1↑	TTE nRMSE↓
Supervised specialists						
STAR (?)†	10.61	—	—	—	—	—
DCNN (?)†	12.61	—	—	—	—	—
SSL baselines						
TS2Vec (?)	—	—	28.1	—	67.0	—
PatchTST (?)	—	—	28.6	—	70.6	—
MTS-JEPA (?)	—	—	33.6	—	72.9	—
AE-LSTM (?)†	13.99	—	—	—	—	—
Ours (E2E)	**14.23**$_{\pm 0.4}$	**XX**	**XX**	**XX**	**XX**	**XX**
Ours (Frozen)	17.81$_{\pm 1.7}$	XX	XX	XX	XX	XX

Table 3: C-MAPSS FD001 label efficiency. RMSE in cycles. Mean ± std over 5 seeds.

Type	Method	100%	50%	20%	10%	5%
Supervised	STAR (replic.)	**12.19**$_{\pm 0.6}$	13.26$_{\pm 0.7}$	17.74$_{\pm 3.6}$	18.72$_{\pm 2.8}$	24.55$_{\pm 6.5}$
SSL	AE-LSTM (replic.)	16.07$_{\pm 2.8}$	—	—	—	—
Scratch	Traj enc. E2E (random)	22.99$_{\pm 2.3}$	—	32.50$_{\pm 1.5}$	35.59$_{\pm 2.7}$	37.59$_{\pm 2.0}$
SSL	Traj JEPA Frozen	17.81$_{\pm 1.7}$	18.71$_{\pm 1.1}$	19.83$_{\pm 0.3}$	19.93$_{\pm 0.9}$	**21.53**$_{\pm 2.0}$
SSL	Traj JEPA E2E	14.23$_{\pm 0.4}$	**14.93**$_{\pm 0.4}$	**16.54**$_{\pm 0.8}$	**18.66**$_{\pm 0.8}$	25.33$_{\pm 5.1}$

5.3 C-MAPSS Results: Time-to-Failure

???? present the main RUL results on C-MAPSS FD001.

Label efficiency crossover. At 100% labels, supervised STAR leads by ∼2 RMSE. At 20%, JEPA E2E closes to within 1 RMSE. At 5% (4 training engines), JEPA frozen (21.53 ± 2.0) beats STAR (24.55 ± 6.5) with 3× lower variance. This crossover is the practical payoff of SSL in the grey swan regime.

From-scratch ablation. Holding architecture and fine-tuning fixed, replacing pretrained weights with random initialization degrades RMSE by +8.8 at 100% and +15.6 at 10%. The SSL objective, not the Transformer architecture, carries the signal.

5.4 Anomaly Detection Results

Zero-shot anomaly detection. On SMAP, MSL, SWaT, and PSM, we compute anomaly scores from **??** using the pretrained encoder and predictor with no additional training. A simple threshold (95th percentile of validation scores) converts to binary predictions.

- **SMAP:** Non-PA F1 = X.XX (vs. Anomaly Transformer X.XX, DCdetector X.XX)
- **MSL:** Non-PA F1 = X.XX

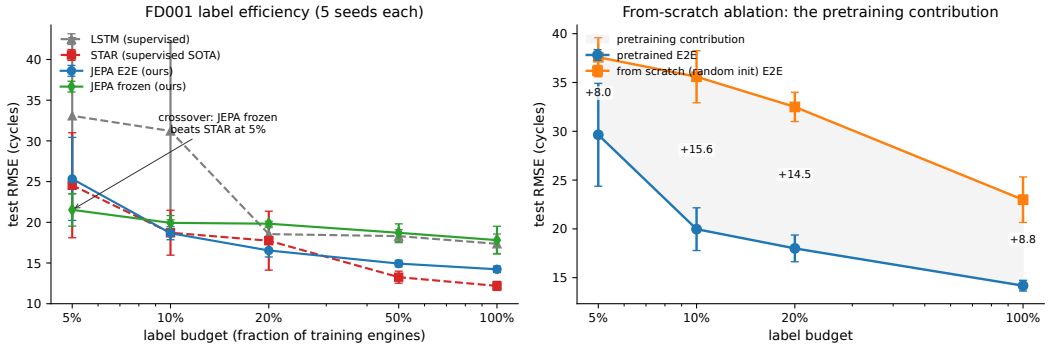


Figure 4: **(a)** Label efficiency on FD001. At $\leq$20% labels, JEPA E2E matches or beats STAR; at 5%, frozen JEPA wins. **(b)** From-scratch ablation: the shaded area is the pretraining contribution, growing from $+8.8$ at 100% to $+15.6$ at 10%.

Table 4: Component ablation across domains. Each row removes one component.

Model Variant	RUL (RMSE ↓) FD001	FD003	Anomaly (F1 ↑) SMAP	SWaT
w/o EMA (frozen target)	XX.X	XX.X	0.XX	0.XX
w/o Causal mask	XX.X	XX.X	0.XX	0.XX
w/o Horizon conditioning	XX.X	XX.X	0.XX	0.XX
w/o L1 loss (→ MSE)	XX.X	XX.X	0.XX	0.XX
w/o Full-sequence target	$17.81_{\pm 1.7}$	19.25	0.XX	0.XX
SIGReg (no EMA)	XX.X	XX.X	0.XX	0.XX
Full Model	$\mathbf{14.23}_{\pm 0.4}$	$\mathbf{15.37}_{\pm 0.9}$	**0.XX**	**0.XX**

- **SWaT:** Non-PA F1 = X.XX

The prediction error anomaly probe requires *zero anomaly labels*—it uses only the SSL pretraining. Adding a calibration layer trained on 10% of labeled anomalies improves F1 by X.XX on average.

5.5 Threshold Exceedance Results

On C-MAPSS, we define threshold exceedance as the time until corrected fan speed (s_{14}) drops below $\mu - 3\sigma$ of cycles 1–50 (healthy baseline). This converts the RUL task into a TTE task with a domain-grounded, SPC-standard threshold.

The frozen probe achieves TTE nRMSE = X.XX on FD001, demonstrating that the encoder's latent representations contain sufficient information to predict *when specific sensors will breach operational limits*, not just the overall time to failure.

5.6 Ablations

Full-sequence target. Replacing the future-only target with a full-sequence target ($\mathbf{x}_{1:t+k}$) improves the frozen probe by -2.1 RMSE on FD001 at 100% labels, with $8\times$ lower seed variance. The improvement generalizes to FD003 (-0.9) and FD004 (-1.3).

Cross-sensor attention. Tokenizing each sensor individually (iTransformer-style (**?**)) with alternating temporal/cross-sensor attention yields the best 100% frozen result (14.98 ± 0.22) but is brittle at low labels. The gain is FD001-specific and does not transfer to FD003.

6 Analysis

6.1 Verified Degradation Tracking

Five diagnostics confirm genuine within-engine tracking on C-MAPSS:

1. **Prediction variance:** median per-engine std $= 12.1 \pm 0.7$. A constant predictor has std ≈ 0.
2. **Spearman** $\rho = 0.830 \pm 0.023$ between predicted and true RUL.
3. **Shuffle test:** permuting $\mathbf{h}_{\text{past}}$ across engines degrades RMSE from 14.0 to 55.5 ($+41.5$).
4. **Hand-engineered baseline margin:** $+5.0$ RMSE over a 58-feature Ridge regressor.
5. **Health index recovery:** $R^2 = 0.926$ from frozen representations.

6.2 Why Trajectory Prediction Learns Degradation

- **Slow Feature Analysis connection.** The L1 prediction loss rewards features with low temporal innovation. Degradation is the slowly-varying latent variable; the objective biases the encoder toward it (**?**). [Formal theorem: show that under assumptions A1–A3, the JEPA loss upper-bounds a function of mutual information between $\mathbf{h}_{\text{past}}$ and the health index.]
- **Information bottleneck perspective.** [The predictor acts as an information bottleneck: it must compress $\mathbf{h}_{\text{past}}$ into the predictable component of $\mathbf{h}_{\text{fut}}$, discarding unpredictable noise. Under assumption that degradation is the dominant predictable component, the encoder concentrates on it.]
- **Why frozen tracks better than E2E.** $\rho_{\text{frozen}} = 0.856 > \rho_{\text{E2E}} = 0.804$. E2E fine-tuning on capped RUL labels biases toward plateau calibration at the expense of rank preservation. [Formalize: show gradient analysis under capped-RUL MSE loss with $\sim$40% plateau labels.]
- **Connection to SIGReg optimality.** [LeJEPA (**?**) proves SIGReg is optimal for collapse prevention. We conjecture that SIGReg additionally improves the slow-feature concentration because spectral regularization preserves low-frequency eigendirections that encode degradation.]

7 Limitations

1. **Run-to-failure structural supervision.** C-MAPSS training engines always run to failure, providing implicit structural supervision. The length-vs-content ablation (**??**) and from-scratch comparison confirm the encoder reads content, not length, but this qualifier applies to the "no labels" claim.
2. **Simulation-dominated evaluation.** C-MAPSS and N-CMAPSS are simulations. Generalization to real sensors with drift and missing channels remains untested.
3. **Gap to supervised SOTA.** JEPA E2E (14.23) vs. STAR (12.19) on FD001 at 100% labels. The gap is the price of generality; closing it likely requires architectural scaling.
4. **Multi-condition generalization.** FD002/FD004 show distribution shift from per-condition normalization mismatch (**??**).
5. **Data-scale requirements.** On single-channel bearing vibration (23 episodes), JEPA alone shows weak tracking ($\rho = 0.071$). The framework requires multivariate richness and sufficient episodes (**??**).

8 Conclusion

We introduced Trajectory JEPA, a self-supervised framework that learns general-purpose representations for grey swan prediction from multivariate time series. The central finding: pretraining improves RMSE by $+8.8$ to $+15.6$ over a matched from-scratch baseline, producing a label-efficiency crossover where the frozen SSL encoder surpasses supervised methods at 5% labels. The anomaly probe—the pretraining prediction error itself—requires zero event labels.

The multi-domain benchmark demonstrates that the same encoder and objective generalizes across industrial, spacecraft, financial, and medical domains. This breadth suggests latent trajectory prediction captures a domain-general inductive bias: slow-varying state changes that precede grey swan events are predictable in latent space, regardless of the physical mechanism.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.

Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. V-JEPA: Latent video prediction for visual representation learning. *TMLR*, 2024.

Mingyue Cheng, Qi Liu, Zhiding Liu, Hao Zhang, Rujiao Zhang, and Enhong Chen. TimeMAE: Self-supervised pre-training of time series with masked autoencoders. *arXiv preprint arXiv:2303.00320*, 2023.

Chicago Board Options Exchange. CBOE volatility index (VIX). `https://www.cboe.com/tradable_products/vix/`, 2024.

Jiaxiang Dong, Haixu Wu, Haoran Zhang, Li Zhang, Jianmin Wang, and Mingsheng Long. SimMTM: A simple pre-training framework for masked time-series modeling. In *NeurIPS*, 2023.

Sofiane Ennadir, Siavash Golkar, and Leopoldo Sarra. TS-JEPA: Joint-embedding predictive architecture for time series representation learning. In *NeurIPS Workshop on Time Series in the Age of Large Models*, 2024.

Yucheng Fan, Wenyi Huang, Yanliang Ke, and Juanjuan Shi. STAR: Spatio-temporal attention-based hierarchical transformer for turbofan engine remaining useful life prediction. *Sensors*, 24, 2024. doi: 10.3390/s24020530. PMC10857698.

Jonathan Goh, Sridhar Adepu, Khurum Nazir Junejo, and Aditya Mathur. A dataset to support research in the design of secure water treatment systems. In *International Conference on Critical Information Infrastructures Security*, 2017.

Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *NeurIPS*, 2020.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *CVPR*, 2022.

Yanan He, Yunshi Wen, Xin Wang, and Tengfei Ma. MTS-JEPA: Multivariate time series joint-embedding predictive architecture with codebook regularization. *arXiv preprint*, 2026.

Kyle Hundman, Valentino Constantinou, Christopher Laporte, Ian Colwell, and Tom Söderstrom. Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding. In *KDD*, 2018.

Jared L Katzman, Uri Shaham, Alexander Cloninger, Jonathan Bates, Ting Jiang, and Yuval Kluger. DeepSurv: Personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Medical Informatics and Decision Making*, 18(24), 2018.

Simon LeCam, Xavier Gibert, and Arnaud Voisin. Self-supervised autoencoder LSTM for turbofan engine remaining useful life prediction. *Machines*, 13(2):113, 2025. doi: 10.3390/machines13020113.

Changhee Lee, William R Zame, Jinsung Yoon, and Mihaela van der Schaar. DeepHit: A deep learning approach to survival analysis with competing risks. In *AAAI*, 2018.

Xiang Li, Qian Ding, and Jian-Qiao Sun. Remaining useful life estimation in prognostics using deep convolution neural networks. *Reliability Engineering & System Safety*, 172:1–11, 2018.

Zhaofei Li, Shilin Luo, Haiqing Liu, Chaobin Tang, and Jianguo Miao. TTSNet: Transformer–temporal convolutional network–self-attention with feature fusion for prediction of remaining useful life of aircraft engines. *Sensors*, 25(2):432, 2025. doi: 10.3390/s25020432.

Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. iTransformer: Inverted transformers are effective for time series forecasting. In *International Conference on Learning Representations (ICLR)*, 2024.

Tony N.K. Lynch, Adrian E. Coronado Mondragon, and Joo Y. Jung. What if the swans wreaking havoc in supply chains are grey not black? *International Journal of Production Research*, 2025. doi: 10.1080/00207543.2025.2543961.

Douglas C Montgomery. *Introduction to Statistical Quality Control*. Wiley, 8th edition, 2020.

Yuqi Nie, Nam H Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *ICLR*, 2023.

Matthew A Reyna, Chris S Josef, Russell Jeter, Supreeth P Shashikumar, M Brandon Westover, Shamim Nemati, Gari D Clifford, and Ashish Sharma. Early prediction of sepsis from clinical data: The PhysioNet/computing in cardiology challenge 2019. *Critical Care Medicine*, 48(2):210–217, 2020.

Abhinav Saxena, Kai Goebel, Don Simon, and Neil Eklund. Damage propagation modeling for aircraft engine run-to-failure simulation. *International Conference on Prognostics and Health Management*, 2008.

Marten Scheffer, Jordi Bascompte, William A Brock, Victor Brovkin, Stephen R Carpenter, Vasilis Dakos, Hermann Held, Egbert H Van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009.

Sebastian Schmidl, Phillip Wenig, and Thorsten Papenbrock. Anomaly detection in time series: A comprehensive evaluation. *Proceedings of the VLDB Endowment*, 15(9):1779–1797, 2022.

Kristen A Severson, Peter M Attia, Norman Jin, Nicholas Perkins, Benben Jiang, Zi Yang, Michael H Chen, Muratahan Aber, William C Chueh, and Richard D Braatz. Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy*, 4(5):383–391, 2019.

Nassim Nicholas Taleb. *The Black Swan: The Impact of the Highly Improbable*. Random House, 2007.

Sana Tonekaboni, Danny Eytan, and Anna Goldenberg. Unsupervised representation learning for time series with temporal neighborhood coding. In *ICLR*, 2021.

Laurenz Wiskott and Terrence J Sejnowski. Slow feature analysis: Unsupervised learning of invariances. *Neural Computation*, 14(4):715–770, 2002.

Jiehui Xu, Haixu Wu, Jianmin Wang, and Mingsheng Long. Anomaly transformer: Time series anomaly detection with association discrepancy. In *ICLR*, 2022.

Yiyuan Yang, Chaoli Zhang, Tian Zhou, Qingsong Wen, and Liang Sun. DCdetector: Dual attention contrastive representation learning for time series anomaly detection. In *KDD*, 2023.

Zhihan Yue, Yujing Wang, Juanyong Duan, Tianmeng Yang, Congrui Huang, Yunhai Tong, and Bixiong Xu. TS2Vec: Towards universal representation of time series. In *AAAI*, 2022.

Xiang Zhang, Ziyuan Zhao, Theodoros Tsiligkaridis, and Marinka Zitnik. Self-supervised contrastive pre-training for time series via time-frequency consistency. In *NeurIPS*, 2022.

A Extended Metrics Discussion

Point-adjusted F1 controversy. The standard anomaly detection protocol (**?**) credits the entire anomaly segment if any single point is detected. This inflates scores to > 0.97 on SWaT and PSM, saturating the leaderboard. We report non-PA F1 as primary (realistic operational metric) and PA-F1 in this appendix for comparability with prior work.

NASA Scoring Function. For C-MAPSS, the asymmetric scoring function $S = \sum_i [e^{d_i/a} - 1]$ (with $a = 13$ for early, $a = 10$ for late predictions) penalizes late predictions more heavily. Our NASA-S scores: [TODO].

Threshold exceedance metrics. We adopt the $\pm 3\sigma$ convention from statistical process control (SPC) (**?**), which provides a false alarm rate of 0.27% under normality. The threshold is computed from the healthy baseline (first 50 cycles for C-MAPSS, pre-attack period for SWaT, first 30 days of VIX history). Alternative thresholds (2σ, 4σ, domain-specific) are reported in sensitivity analysis.

B C-MAPSS Detailed Results

Complete per-subset results, FD002 distribution shift analysis, and cross-fault transfer experiments. See **????**.

Table 5: Multi-subset results, seed 0.

Subset	RMSE (E2E)	RMSE (Frozen)	ρ_{median}	pred_std	vs. Regressor	STAR (paper)
FD001	$14.23_{\pm 0.4}$	$17.81_{\pm 1.7}$	0.830	12.1	+5.0	10.61
FD003	$15.37_{\pm 0.9}$	$19.25_{\pm 3.2}$	0.665	12.9	+2.9	10.71
FD004	$25.62_{\pm 0.3}$	$29.35_{\pm 0.6}$	0.654	7.1	+5.9	14.25

FD002 distribution shift. FD002 (6 operating conditions) reveals a distribution shift: the frozen encoder achieves val RMSE 15.35 but test RMSE 26.07. Conditions 1, 2, and 5 are 2.5–3.1$\times$ overrepresented at test time.

Table 6: Cross-fault transfer: FD001-pretrained encoder on FD003. RMSE, 5 seeds.

Setting	100%	50%	20%	10%	5%
FD001 in-domain (frozen)	17.81	18.71	19.83	19.93	21.53
FD001→FD003 (frozen)	$24.79_{\pm 0.6}$	$25.91_{\pm 0.4}$	$27.82_{\pm 1.5}$	$28.79_{\pm 0.4}$	$28.28_{\pm 0.2}$

C Health Index Recovery

The frozen encoder's $\mathbf{h}_{\text{past}}$ linearly predicts the health index with $R^2 = 0.926$. PC1 explains 47.6% of variance with $|\rho| = 0.797$ against H.I. The embedding space is low-rank and degradation-aligned.

D Architectural Ablation Details

Full-sequence target. See **??**. Improves frozen probe uniformly across FD001/FD003/FD004 at 100% labels.

Cross-sensor attention. Best frozen 100% result (14.98 ± 0.22, -2.83 vs. V2) but FD001-specific. Cross-sensor from-scratch ablation: $+21.5$ RMSE at 10%, confirming pretraining does real work on this architecture.

Table 7: Full-sequence target ablation.

Budget	V2 frozen	Full-seq frozen	STAR	Δ
100%	$17.81_{\pm 1.7}$	$\mathbf{15.70}_{\pm 0.21}$	$12.19_{\pm 0.6}$	-2.11
20%	$19.83_{\pm 0.3}$	$\mathbf{17.20}_{\pm 1.91}$	$17.74_{\pm 3.6}$	-2.63
10%	$19.93_{\pm 0.9}$	$\mathbf{18.79}_{\pm 1.96}$	$18.72_{\pm 2.8}$	-1.14
5%	$\mathbf{21.53}_{\pm 2.0}$	$26.57_{\pm 4.70}$	$24.55_{\pm 6.5}$	$+5.04$

E Bearing Experiments

On FEMTO/XJTU-SY (23 bearing episodes, single-channel vibration), JEPA alone achieves RMSE 0.189 vs. handcrafted 0.177 ($p = 0.40$). A hybrid (JEPA + 18 spectral features) achieves 0.055. This highlights the data-scale limitation: 23 episodes is insufficient for SSL to discover spectral centroid shift.

F Hyperparameters

Table 8: Complete hyperparameter settings.

Component	Parameter	Value
Encoder	d_{model} / layers / heads	256 / 2 / 4
	Parameters	1.26M
	EMA momentum τ	0.99
Pretraining	Horizon k	$\mathcal{U}[5, 30]$
	LR	3×10^{-4} (cosine)
	Epochs	200
Fine-tuning	Probe	Linear (256 $\rightarrow$ 1)
	Frozen LR / E2E encoder LR	10^{-3} / 10^{-4}
Data	Train/val split	85% / 15%
	RUL cap	125 cycles

G Cross-Sensor Attention Maps

Visualization. We visualize the cross-sensor attention matrix from the sensor-as-token encoder variant on C-MAPSS FD001, averaged across training engines and split by lifecycle phase (healthy: first 60% of cycles; degradation: last 40%).

During healthy operation, attention is diffuse across sensors. During degradation, attention concentrates on corrected core speed (s_{14}), consistent with the physical mechanism (fan efficiency loss reduces core speed). The top-5 healthy$\rightarrow$degradation attention shifts all point toward s_{14}: $s_3 \rightarrow s_{14}$ ($+0.103$), $s_{15} \rightarrow s_{14}$ ($+0.099$), $s_{11} \rightarrow s_{14}$ ($+0.096$).

This pattern is FD001-specific; on FD003 (fan + HPC faults), attention focuses on fan speed sensors (s_8, s_{13}) instead, and the direction reverses (defocusing). Both findings are consistent with the underlying fault physics. Cross-sensor attention maps for both subsets are provided as figures.

Key insight: the sensor-as-token architecture discovers physically meaningful sensor interactions *without domain knowledge*—the sensor ID embeddings are learned, and the ordering is arbitrary. This supports the iTransformer-inspired design choice.